

CVE Database Automation using NLP

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Introduction

- Cybersecurity vulnerabilities are increasing rapidly every year.
- CVE databases contain large volumes of vulnerability records.
- Manual vulnerability analysis is time-consuming and difficult.
- NLP automates extraction and classification of vulnerabilities.
- Transformer models improve contextual understanding.
- Intelligent automation improves cybersecurity systems.

Goal: Develop an intelligent NLP-based CVE analysis system for automated cybersecurity vulnerability management.

- Cyber attacks are increasing exponentially worldwide.
- Organizations require intelligent vulnerability management systems.
- Large CVE datasets are difficult to process manually.
- NLP enables semantic understanding of cybersecurity text.
- Automated severity prediction improves risk assessment.
- Trend analysis helps predict future cybersecurity threats.

Need: Automated extraction and intelligent analysis of cybersecurity vulnerabilities.

Research Gap

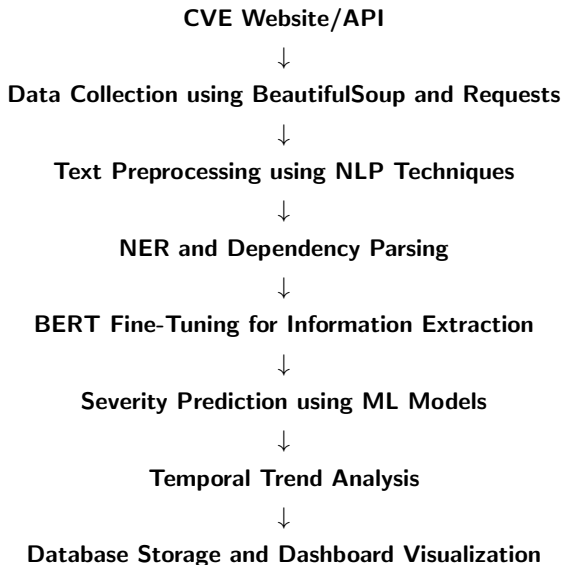
- Traditional databases provide only static storage.
- Rule-based methods fail for complex attack descriptions.
- Classical ML models lack contextual understanding.
- Limited systems use transformer-based NLP models.
- Existing research lacks temporal vulnerability analysis.
- Few intelligent end-to-end CVE automation systems exist.

Research Gap: Lack of intelligent Transformer-based cybersecurity automation systems.

Problem Statement

Research Problem

How can NLP and Transformer-based models automate the extraction, classification, severity prediction, and trend analysis of cybersecurity vulnerabilities from CVE databases to improve cybersecurity intelligence and vulnerability management systems?



Trend Analytics Dashboard

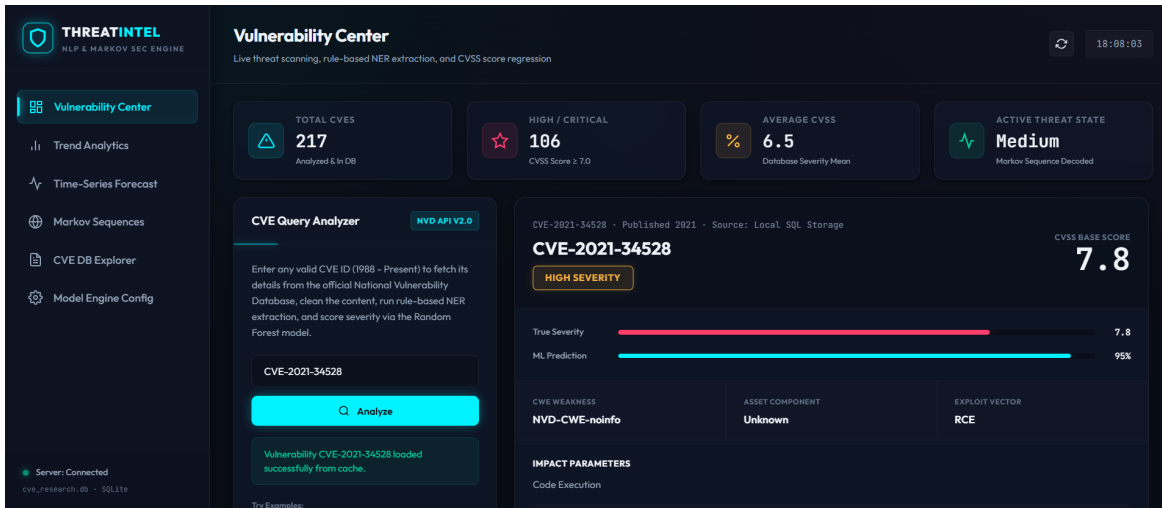


Figure: Trend Analytics Dashboard showing exploit vector analysis and affected components.

CVSS Distribution Dashboard



Figure: CVSS Base Score Distribution Dashboard.

Time-Series Forecast Dashboard



Figure: Vulnerability Volume Forecast using ARIMA and Regression Models.

Markov Threat Sequence Dashboard



Figure: Hidden Markov Model Threat State Clustering Visualization.

SQLite Vulnerability Vault

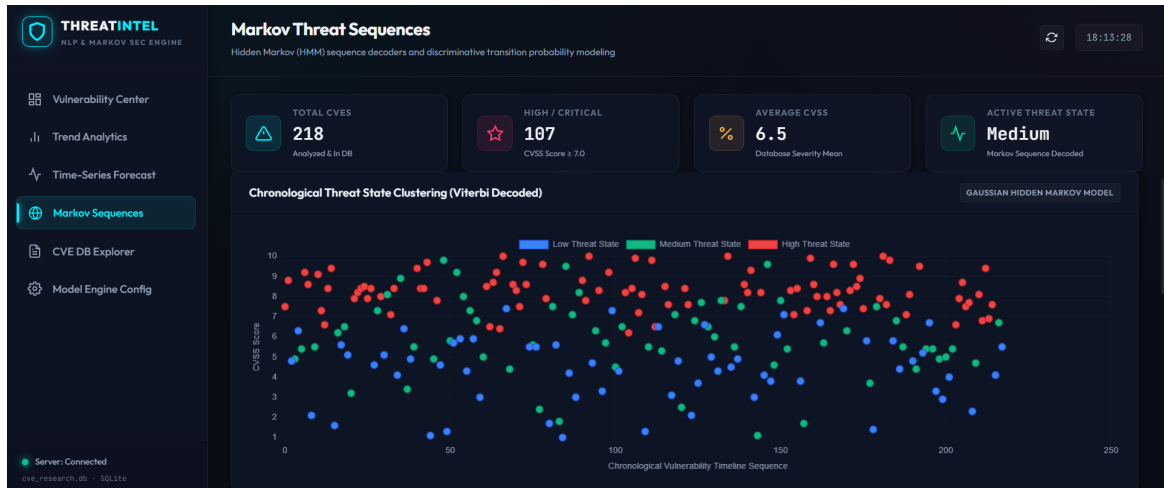


Figure: SQLite Vulnerability Database Explorer Interface.

Model Engine Configuration

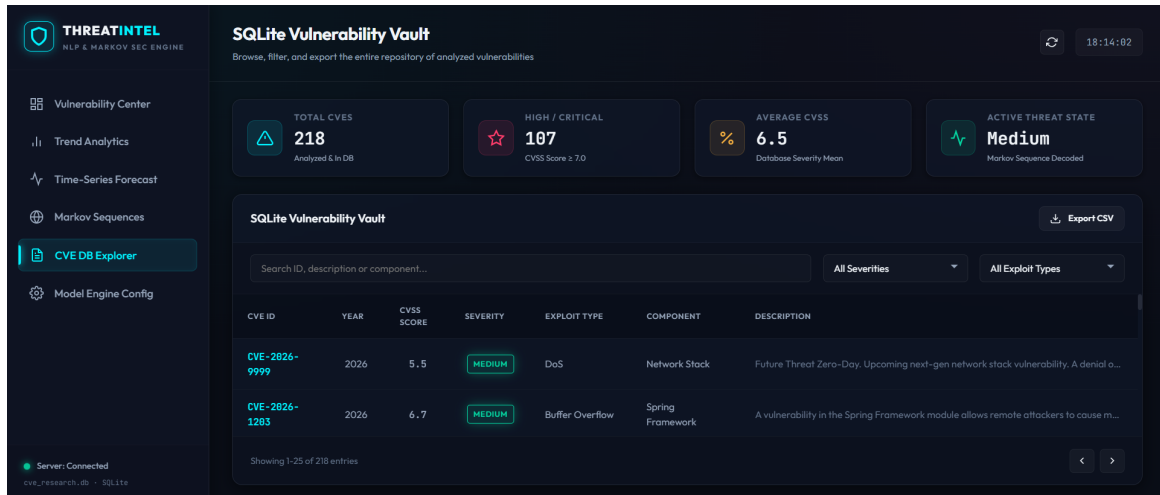


Figure: Machine Learning Model Engine Configuration Dashboard.

Vulnerability Center Dashboard

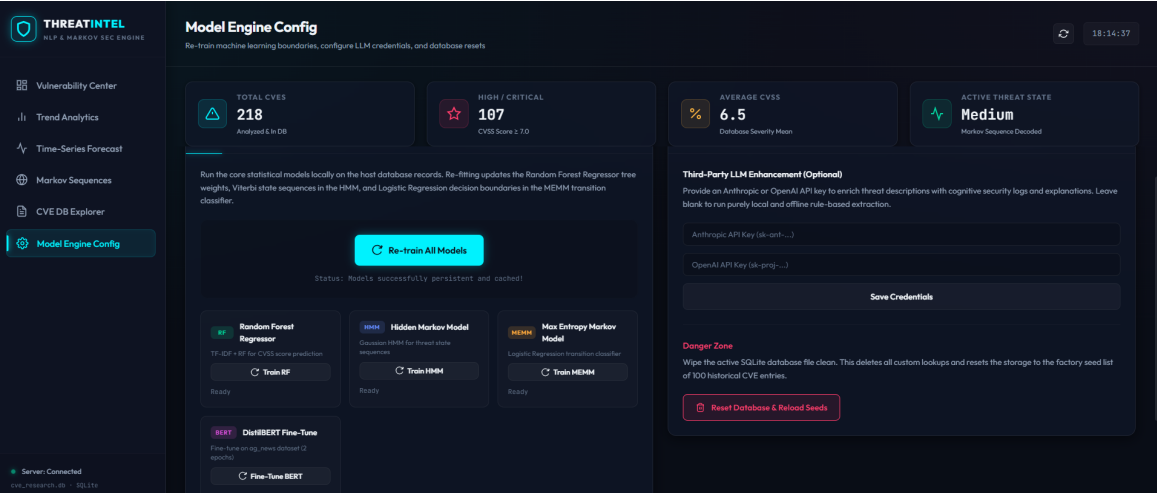


Figure: CVE Query Analyzer and Vulnerability Intelligence Dashboard.

Conclusion

- NLP significantly improves cybersecurity automation systems.
- BERT enhances vulnerability extraction and classification.
- Automated severity prediction improves risk prioritization.
- Trend analysis supports proactive cybersecurity planning.
- Intelligent systems reduce manual workload effectively.
- The project strengthens cybersecurity intelligence capabilities.

Conclusion: Transformer-based NLP models provide intelligent and scalable cybersecurity vulnerability analysis solutions.

Future Scope

- Real-time CVE monitoring and alert generation.
- Integration with SIEM and SOC platforms.
- Explainable AI for transparent prediction systems.
- Multi-language vulnerability analysis support.
- Graph-based cybersecurity threat intelligence systems.
- Cloud deployment for scalable cybersecurity analytics.

Future Enhancement: Develop fully autonomous AI-driven cybersecurity intelligence systems.

Thank You

Questions and Discussion